1883

June 14, 2023

Exam Time: 60 minutes. Solve all questions (25x4) and show all steps

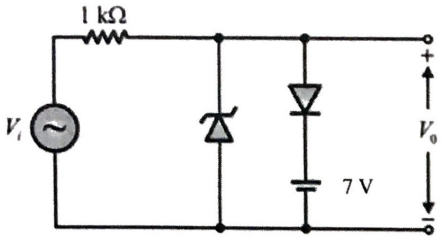
Name	ID	Mark
KEY	α=**Sum** of the indicated digits of your ID number	/100

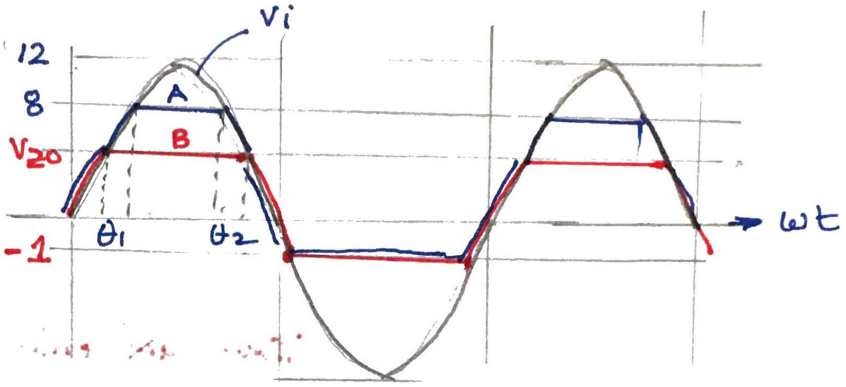
1) For the circuit shown, the diodes are modeled by the constant voltage drop of 1.0 V when conducting in forward mode. The zener diode has $V_{Z0}=\alpha$ V and $r_Z=0\ \Omega$.

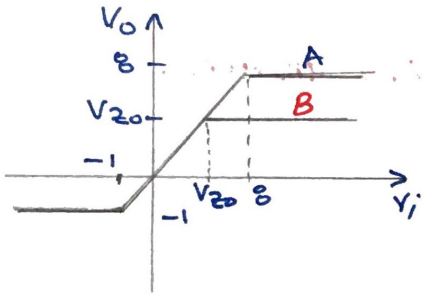
a) Draw the output voltage waveform if the input is a sinusoidal signal of amplitude 12 V.

b) Draw the transfer characteristics of the circuit.

Label all points on the curves. Explain your work.





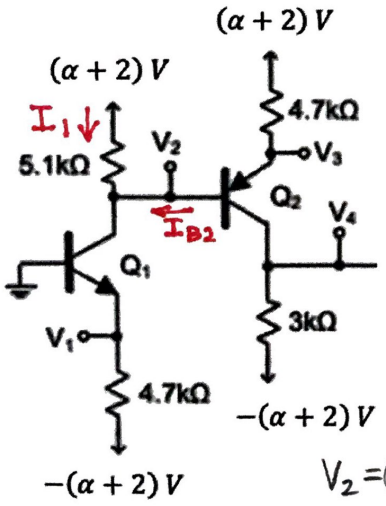


$\alpha > 8$: Curve A : $\theta_1 = \sin^{-1}\left(\frac{8}{12}\right)$; $\theta_2 = \pi - \theta_1$

$\alpha \leq 8$: Curve B : $\theta_1 = \sin^{-1}\left(\frac{\alpha}{12}\right)$; $\theta_2 = \pi - \theta_1$

2) In the DC circuit given below, all transistors have $\beta=100$. Find the voltages V_1 to V_4. Do not ignore the base current in any of the transistors. Show all your work.

V_1	V_2	V_3	V_4



$V_{B1} = 0$

$V_1 = V_{E_1} = -0.7$V

$I_{E_1} = \frac{-0.7 + (\alpha+2)}{4.7\text{K}}$

$I_{C_1} = 100 I_{E_1}/101$

$I_1 = I_{C_1} - I_{B2}$

$V_2 = (\alpha+2) - 5.1 I_1$

Also

$V_2 = V_3 - 0.7$

$V_2 = (\alpha+2) - 4.7 I_{E_2} - 0.7$

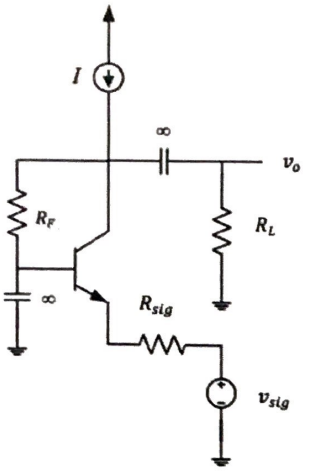
Equate the two V_2 equations and put $I_{E_2} = (100+1) I_{B2}$

$$\Rightarrow I_{B2} = \frac{5.1 I_{C_1} - 0.7}{4.7(101) + 5.1}$$

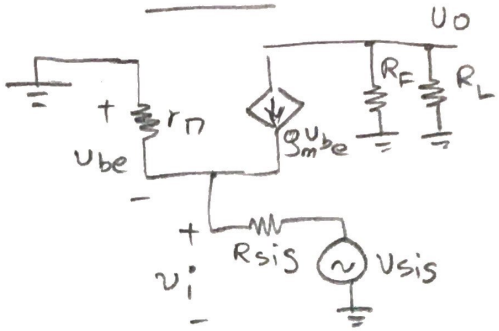
$V_3 = (\alpha+2) - 4.7(101) I_{B2}$

$V_4 = (101) I_{B2} (3) - (\alpha+2)$

3) Analyze the amplifier circuit below. Produce the small-signal equivalent circuit using:
the pi-model for $\alpha \leq 8$
the T-model for $\alpha \geq 9$.
Ignoring r_o, write expressions for (a) the amplifier voltage gain (b) the input resistance (c) the output resistance.



$\alpha \leq 8$

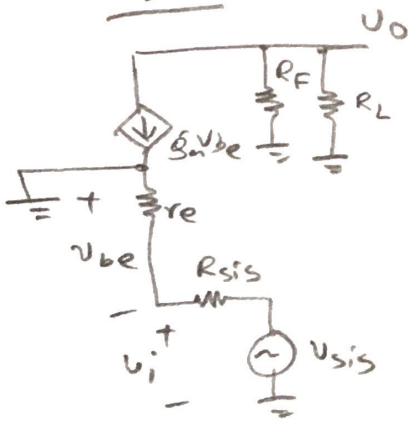


$v_i = -v_{be}$

$v_o = -g_m v_{be} (R_F // R_L)$

$\frac{v_o}{v_i} = g_m (R_F // R_L)$

$\alpha \geq 9$



$R_i = r_e$

$R_o = R_F$

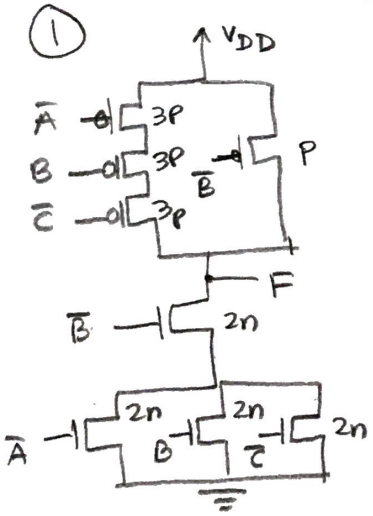
These are only final answers!

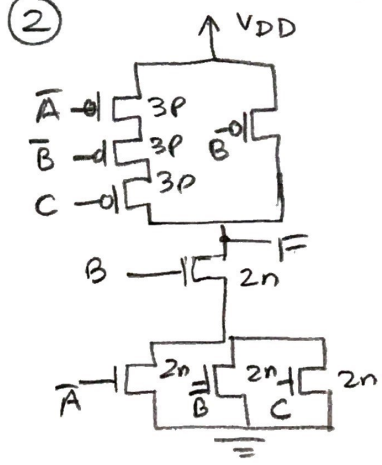
4) Implement the given logical function using CMOS logic. Perform transistor sizing such that the performance of the circuit matches the basic CMOS inverter with transistor size n and p. Assume that the input and their complements are available.

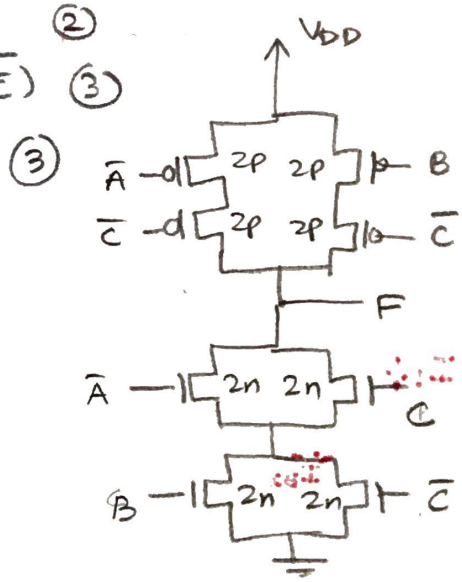
$1 \leq \alpha \leq 7$: $F(A,B,C) = B + A\bar{B}C = \overline{\bar{B}(\bar{A}+B+\bar{C})}$ (1)

$8 \leq \alpha \leq 12$: $F(A,B,C) = AB\bar{C} + \bar{B} = \overline{(\bar{A}+\bar{B}+C)B}$ (2)

$\alpha \geq 13$: $F(A,B,C) = A\bar{C} + \bar{B}C = (\bar{A}+C)(B+\bar{C})$ (3)







MOSFET Current Equations	Triode	$I_D = k\,[2(V_{GS} - V_t)V_{DS} - V_{DS}^2]$	Saturation	$I_D = k\,(V_{GS} - V_t)^2(1 + \lambda V_{DS})$
BJT Current Equations	Active	$I_C = \beta I_B, \quad I_E = (\beta + 1)I_B$	Relations	$I_E = I_C + I_B$
Diode Current Equations	Forward	$I_D = I_s\,(e^{V_D/nV_T} - 1)$	Roots	$x = (-b \pm \sqrt{b^2 - 4ac})/2a$
Amplifier Equations	Gain	$A_v = v_o/v_i, \quad A_i = i_o/i_i$	Resistance	$R_i = v_i/i_i, \quad R_o = v_o/i_o$